① $\mu = np$

② $\sigma^2 = npq$

③ $\sigma = \sqrt{npq}$

ex

X	P(X)
0	$3C_0 (\frac{1}{4})^0 (\frac{3}{4})^3 = \frac{27}{64}$
1	$3C_1 (\frac{1}{4})^1 (\frac{3}{4})^2 = \frac{27}{64}$
2	$3C_2 (\frac{1}{4})^2 (\frac{3}{4}) = \frac{9}{64}$
3	$3C_3 (\frac{1}{4})^3 = \frac{1}{64}$
total	1

$n = 3$

$P = \frac{1}{4} = 0,25$

$q = 0,75 = \frac{3}{4}$

① in the previous find mean, variance, σ

② propability of $P(X) \geq 2$ at least

③ $P(X) \leq 2$ atmost

Suppose that X is normal with $\mu = 10$, $\sigma = 2$, Fid. $P(15 < X < 10)$

Find $P(12 < X < 17)$

Find area between $0 < X < 20$

CamScanner

Compute that

$$P(X \leq 20) = P(X=0) + P(X=1) + P(X=2) \cdots P(X=12)$$

$$= \sum_{i=0}^{12} \frac{e^{-10} 10^{i}}{i!}$$

assignment

$E(X) = \lambda$

$var(X) = \lambda$

$S.d(X) = \sqrt{\lambda}$

variable $X \longrightarrow x_1, x_2, \ldots, x_n$

(random variable) $Y \longrightarrow Y_1, Y_2, \ldots Y_n$

$\downarrow \quad \downarrow \quad \downarrow$

$P_1 \quad P_2, \ldots P_n$

probability distribution :

* Sample space (coin) : 2^{n} → No. of times

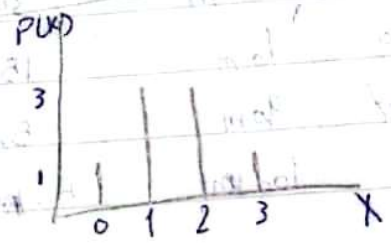
sample space (dice) : 6^{n}

لو رميت coin 3 مرات وعايز ال head

Result	Coin toss 1	2	3	No of head
1	T	T	T	0
2	T	T	H	1
3	T	H	T	1
4	H	T	T	1
5	T	H	H	2
6	H	T	H	2
7	H	H	T	2
8	H	H	H	3

the probability dist of No of heads

No of head X	P(x)
0	$\frac{1}{8}$ = 0.125
1	$\frac{3}{8}$
2	$\frac{3}{8}$
3	$\frac{1}{8}$
total	1



* mean, S.D ← measures

① $0 \leq P(X) \leq 1$

② out comes are mutually exclusive

③ $\sum P(x) = 1$

characteristics of distribution.

random variable
- discrete r.v (counting) (points)
- continous r.v (measuring) (Range)